CE 310

Hypothesis Testing

Week 11 · Session 1 · t-Tests, ANOVA, and Effect Size

University of Arizona · Dr. Wooyoung Jung · Fall 2026

The Pour Is Already in the Ground

90 seconds · accept, or core?

A 900 yd³ mat foundation was placed 28 days ago. The cylinder breaks are in.

Specified f′c	Cylinders	Sample mean	Sample SD
4,000 psi	6	**3,890 psi**	190 psi

- **The contractor** — "110 psi on six cylinders is noise. Accept the pour."
- **The owner's rep** — "The mean is below spec. Reject it — remove and replace."

> Removal and replacement runs **$340,000**, and steel erection starts in 11 days. You are the engineer of record. **Which one is right?**

✅ "Cannot Reject" Is Not "The Concrete Is Fine."

The test answers a narrower question than the one that was asked

One-sample t-test of the six breaks against the 4,000 psi specification:

Quantity	Value
t statistic (5 df)	**–1.42**
p-value (two-tailed)	**0.22**
95% confidence interval on the true mean	**3,690 – 4,090 psi**

At $\alpha = 0.05$ you **cannot** conclude the concrete is below spec. But read that interval: with six cylinders, the data are equally consistent with 3,700 psi and with 4,090 psi.

> ⚠️ At this sample size the test would miss a **real** 200 psi deficit 45% of the time. The honest recommendation is the one neither engineer wrote: **six cylinders cannot settle this — take cores.**

The Question Behind the Numbers

Contracts let with **3 bidders or fewer** are won at **1.008** times the agency's own estimate. Contracts with **6 or more** are won at **0.832**.

But how do you know that gap isn't just the luck of which jobs drew a crowd?

Maybe the busy lettings happened to be simple paving jobs that everyone prices tightly. Maybe a different two years of data would show no gap at all.

Hypothesis testing is the formal answer:

- It asks: *what is the probability that sampling noise alone produced a gap this large, if competition made no difference at all?*
- If that probability is very small ($p < 0.05$), the gap is **real** — not noise.

> 💡 On a median contract this is not an abstraction. Seventeen cents on every estimated dollar is the difference between a programme that funds four jobs and one that funds five.

Anatomy of a Hypothesis Test

Every test follows the same five steps — memorize the structure once; apply it to any test.

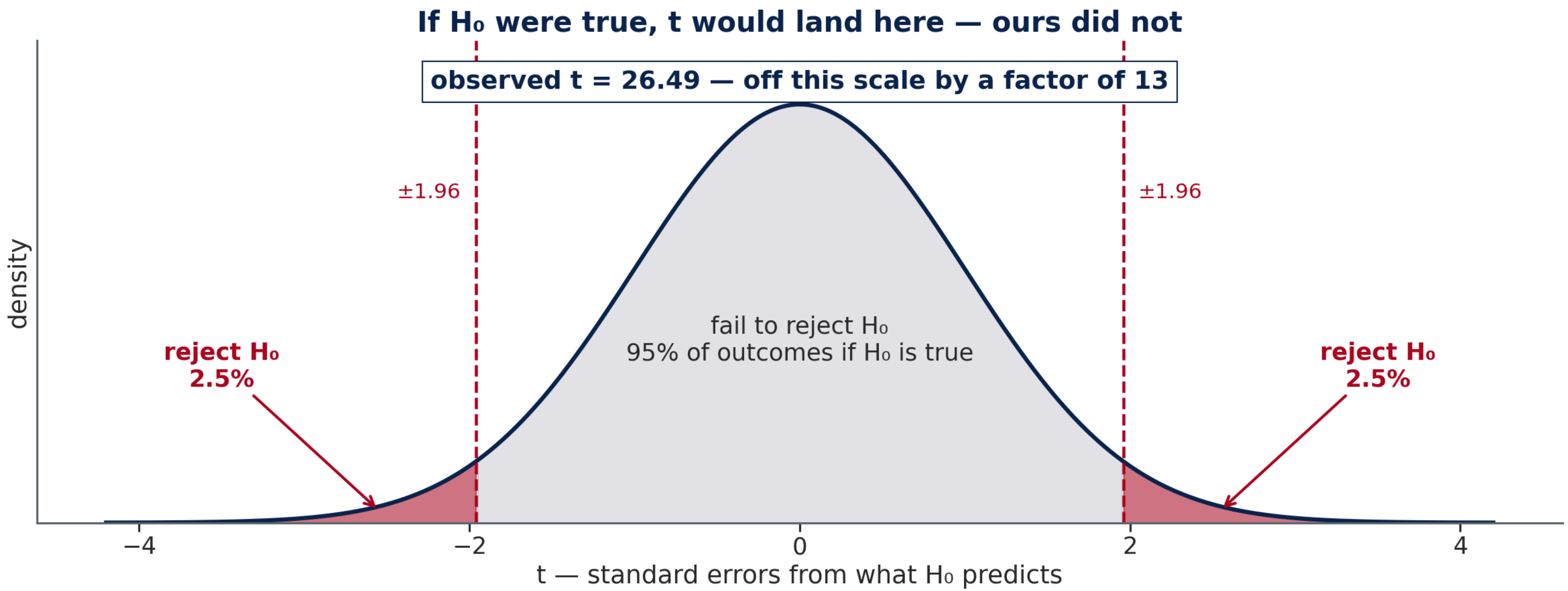
1. **State H_0 (null):** the default "no effect" assumption, always written with $=$. Example: $H_0 : \mu_{occ} = \mu_{unocc}$
2. **State H_1 (alternative):** what you want to show. *Two-tailed* ($\neq$): difference in either direction — most common. *One-tailed* ($>$ or $<$): specific direction, requires prior justification.
3. **Choose α — before seeing the data:** typically 0.05 (5% false-positive rate accepted in advance). Choosing α after seeing the result is p-hacking.

Anatomy of a Hypothesis Test (cont.)

4. **Compute the test statistic:** measures deviation from H_0 in standard-error units. Two groups → t-statistic. Three or more groups → F-ratio.
5. **Interpret the p-value:** probability of a result this extreme or more extreme *if H_0 were true*. $p < \alpha$ → reject H_0.

> "Fail to reject H_0" ≠ "H_0 is true." The data simply lack enough evidence to rule it out — like a "not guilty" verdict that does not mean innocent.

What "If H_0 Were True" Actually Means



Every hypothesis test compares one observed number against the range that number *would* take if H_0 were true. Everything else is bookkeeping.

5 Steps Applied — Thin vs. Deep Competition

The five-step structure

1. $\mathbf{H_0}$ — null assumption (always uses $=$)
2. $\mathbf{H_1}$ — what to show ($\neq$, $>$, or $<$)
3. **α** — false-positive rate, set in advance
4. **Test statistic** — signal ÷ noise in SE units
5. **Decision** — compare p-value to α

Applied to 4,575 awarded contracts

1. $H_0 : \mu_{thin} = \mu_{deep}$ — competition has no effect
2. $H_1 : \mu_{thin} \neq \mu_{deep}$ — two-tailed
3. $\alpha = 0.05$
4. $t \approx 26.5$ (gap of 0.176 ÷ SE ≈ 0.0067)
5. $p \approx 0 < 0.05$ → **Reject $\mathbf{H_0}$**

The same five-step template applies to every test this week: the one-sample test against the engineer's estimate, the one-way ANOVA across districts, and every coefficient in the `P>|t|` regression output.

Two Ways a Hypothesis Test Can Be Wrong

Rejecting or failing to reject H_0 is a decision made under uncertainty — it can be wrong in two distinct ways.

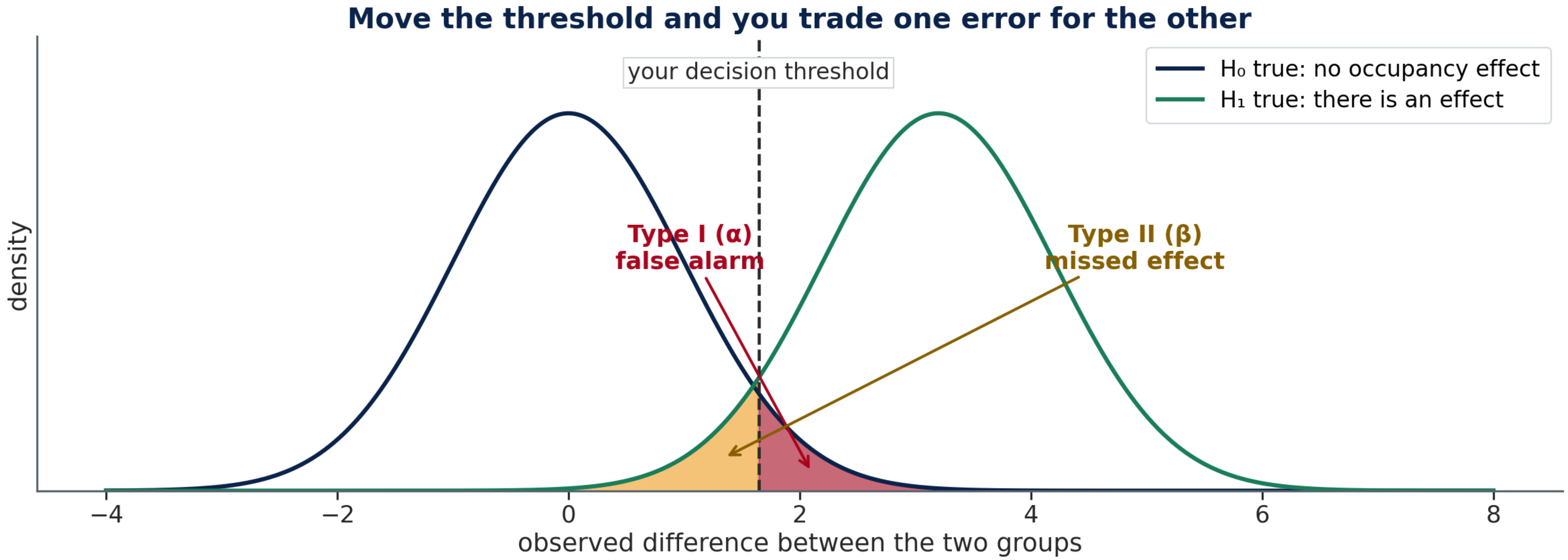
	H_0 is actually true	H_0 is actually false
You reject H_0	**Type I error** (false positive) — probability = α	Correct decision
You fail to reject H_0	Correct decision	**Type II error** (false negative)

Two Ways a Hypothesis Test Can Be Wrong (cont.)

- **Type I error (α):** you conclude there's an effect when there isn't one. This is exactly the "false alarm" rate from Step 3 of the 5-step framework — choosing $\alpha = 0.05$ means you accept a 5% chance of a false positive on any single test.
- **Type II error:** you conclude there's no effect when there actually is one — often from too small a sample or too small a true effect relative to noise.

> 💡 α is not a mistake rate you observe after the fact — it's a false-positive rate you accept in advance. The "26% false alarm rate across 6 t-tests" result later this lecture is Type I error compounding across multiple tests.

The Two Errors Trade Against Each Other



You cannot shrink both at once with fixed data. Moving the threshold left catches more real effects and raises the false-alarm rate; moving it right does the reverse.

The t-Statistic — What It Measures

The t-statistic answers: *how many standard errors is the observed difference from what H_0 predicts?*

$t = (\text{mean}_1 - \text{mean}_2)$ / standard error of the difference

- **Numerator:** the observed difference between group means
- **Denominator:** how much that difference varies due to chance (shrinks as n grows)

Interpreting t:

- $|t|$ close to 0 → difference is within sampling noise → do not reject H_0
- $|t|$ large (> ~2) → difference is many standard errors from zero → reject H_0

For the competition gap: **$t \approx 26.5$** — the 0.176 difference in bid-to-estimate ratio is 26.5 standard errors from zero. Sampling noise cannot produce a t-statistic this large.

Degrees of Freedom — Why n – 1 Again

To turn *t* into a *p*-value, Python compares it against a **t-distribution** — and that curve's shape is set by a single number: the **degrees of freedom**.

Where it comes from

Estimating the mean from your own sample spends one piece of information. Hold the mean fixed and only n – 1 of the values are free to vary — the last one is forced.

It is the same n – 1 you divided by for a sample standard deviation in Week 1.

Test	Degrees of freedom
One-sample *t*	n – 1
Two-sample *t*	$n_1 + n_2 - 2$
One-way ANOVA	k – 1 and N – k

Small df → heavier tails, larger *t* needed

Large df → converges on the Normal

💡 Two means estimated, two degrees of freedom spent — hence the $n_1 + n_2 - 2$ in the pooled SD.

Two-Sample t-Test — Thin vs. Deep Competition

H_0: mean winning ratio with ≤ 3 bidders = mean winning ratio with ≥ 6
H_1: the means are not equal · **α = 0.05**

A **two-sample *t*-test** compares the two means and returns the *t* statistic and its *p*-value.

Thin competition: **1.008** (n = 1,411) · Deep competition: **0.832** (n = 1,783)

Expected: $t \approx 26.5$ · $p \approx 5 \times 10^{-138}$ → **Reject H_0**

The p-value is effectively zero. With t = 26.5, the gap is 26.5 standard errors from zero — sampling noise cannot explain a difference this large.

> 💡 Use `equal_var=False`. Welch's t-test does not assume the two groups share a variance, and you rarely know that they do.

A Caution: These Data Are Not Independent

t = 26.5 assumes every observation is independent. **Bid tabulations are not.**

Clustering: the 4,369 excavation bids come from only **893 projects** — about five each. Contractors bidding the same job share a site, a haul distance and a schedule, so their prices move together.

What this means for the p-value:

- **Effective n** is far below the row count
- Closer to the number of *projects* than of *bids*
- The true t-statistic is smaller than the naive one

Does the conclusion change here?

No — the gap is large enough to survive a severe correction. For a marginal result, clustering can flip a conclusion.

⚠️ Every p-value this week is a naive one. Say so when you report it.

One-Sample t-Test — Comparing to a Benchmark

When you test whether a sample mean equals a **known reference value** rather than another sample, use the one-sample t-test.

Example: every project carries an **engineer's estimate**, prepared by the agency before bids are opened. A bid-to-estimate ratio of 1.0 means the bid landed exactly on it.

H_0: μ_ratio = 1.0 — bids track the estimate
H_1: μ_ratio ≠ 1.0

For roadway excavation the mean ratio is **1.207** and t ≈ **18.7**, so H_0 is rejected overwhelmingly.

But the **median ratio is exactly 1.000**, and only **40.9%** of bids exceed the estimate.

> ⚠️ Both statements are true. This is the central tension of Week 11, and the in-class exercise asks you to resolve it.

Quick Check — Big Sample, Big Problem?

60 seconds · pick A or B

Three comparisons from this week's data, all with $p < 0.05$:

- Winning ratio: thin vs. deep competition — **d = 0.96**
- Unit price: construction vs. maintenance contracts — **d = 0.80**
- Quantity bid: winners vs. losers — **d = 0.10**

All three are "statistically significant." Which interpretation is correct?

- **A)** All three matter equally for procurement policy — the same p-value means the same importance
- **B)** The p-value tells you the effect is real, not how large it is

✅ Answer: B — p-values Don't Measure Size

Statistical significance ≠ practical significance

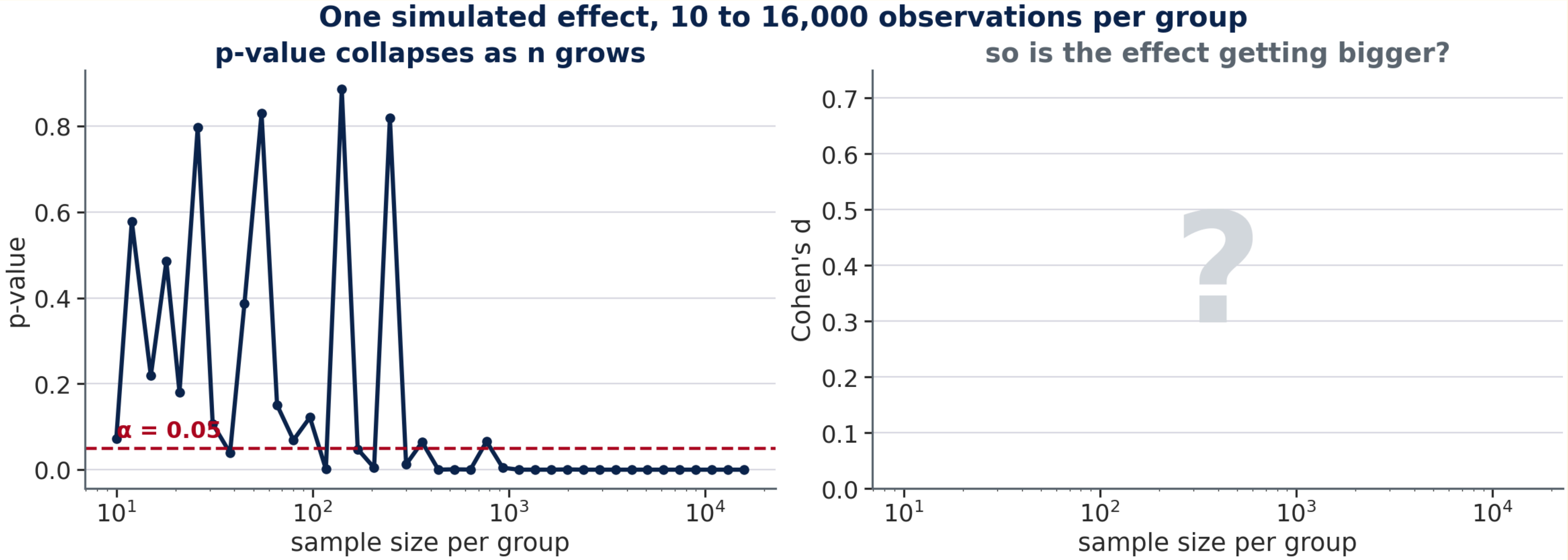
Comparison	p-value	Cohen's d	What to do about it
Thin vs. deep competition	≈ 0	**0.96** large	Change how work is packaged and advertised
Construction vs. maintenance	2×10^{-6}	0.80 medium	Price the two contract types separately
Quantity: winners vs. losers	0.031	0.10 negligible	Real, but far too small to act on

The rule: with thousands of rows, nearly every difference is statistically significant. A significant p-value means the effect is **real** — it does NOT mean the effect is **large enough to matter**.

> ⚠️ The third row is the one that bites. It is significant, and it is nothing. Always report an effect size beside a p-value.

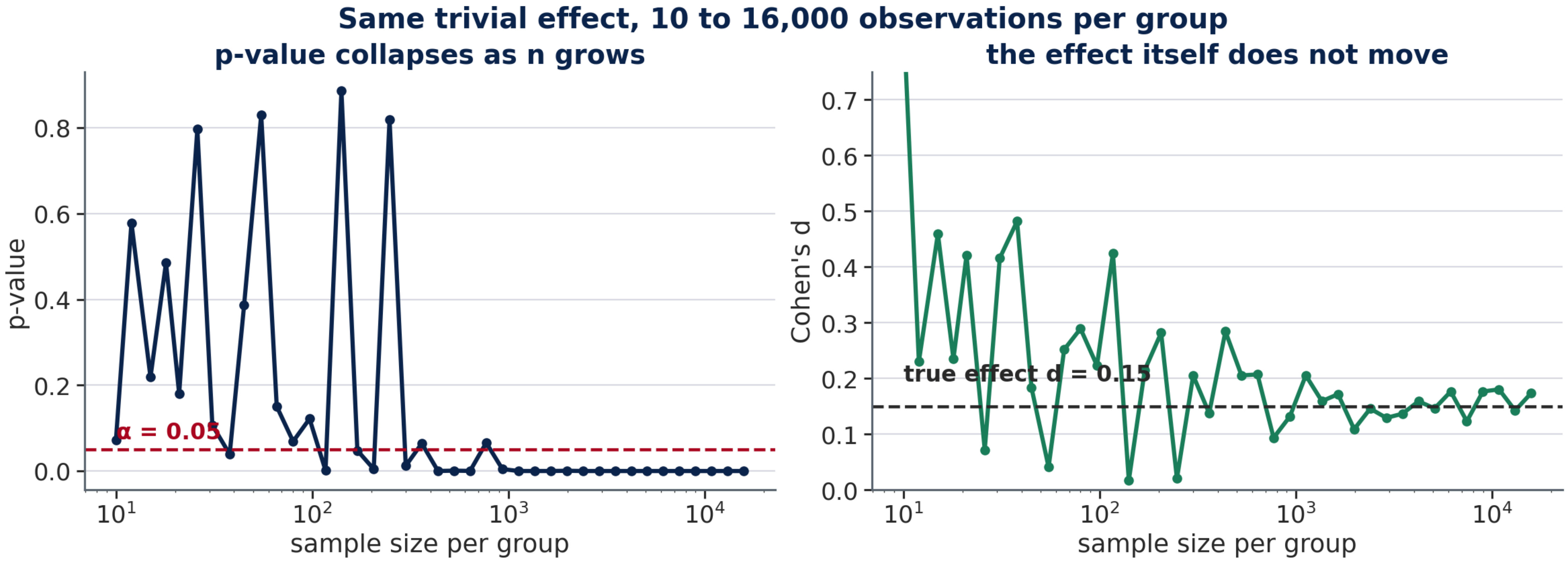
p Keeps Dropping. Is the Effect Getting Bigger?

45 seconds · commit before the next slide



The p-value falls off a cliff. **What is the curve on the right doing — rising, flat, or falling?**

Same Effect, Different n



The effect never changed. Only our ability to detect it did. This is why a p-value alone cannot tell an engineer whether something is worth acting on.

Statistical vs. Practical Significance

A p-value tells you an effect is **real** — not noise. It does not tell you whether the effect is **large enough to matter.**

Two results from this week that pull in opposite directions:

Result	p-value	Cohen's d	Reading
Winning ratio: thin vs. deep	≈ 0	0.96	Significant **and** large
Unit price: winners vs. losers	0.087	–0.08	Neither significant nor large
Bid ÷ estimate vs. 1.0	2×10^{-75}	—	Significant, but the median is 1.000

⚠️ The third row is the trap: a test about a *mean* says nothing about a typical bid when the tail is long.

Effect Size — Cohen's d

Cohen's d measures a difference in standard-deviation units — independent of sample size.

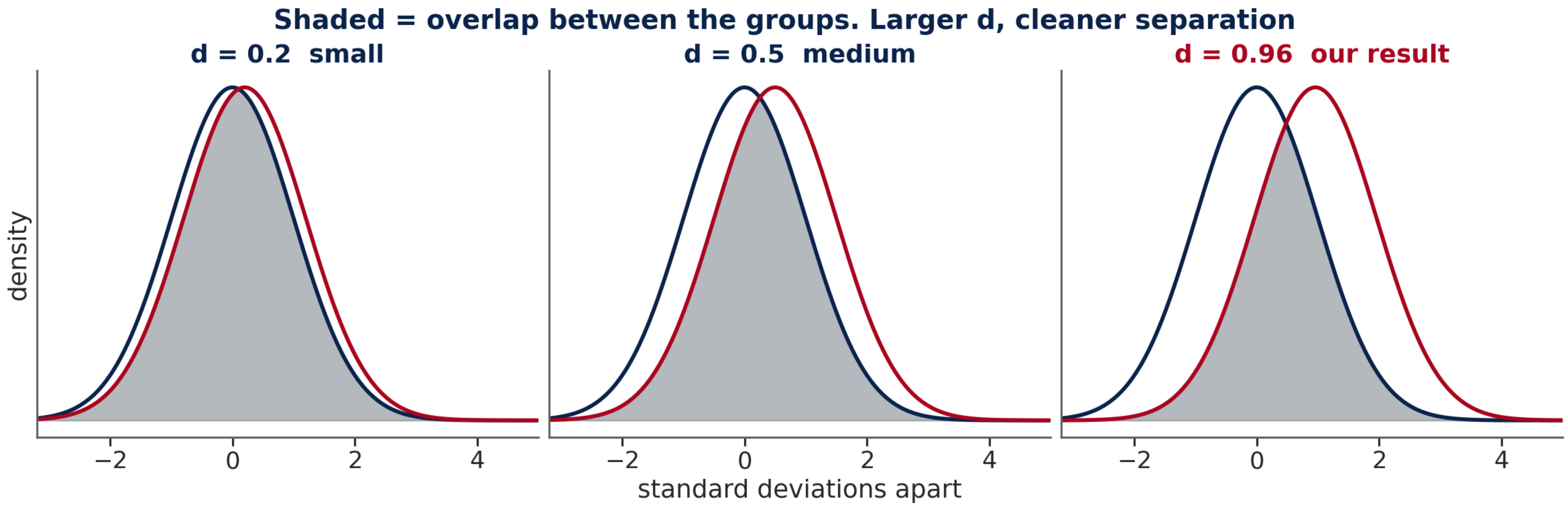
$$d = \frac{\bar{x}_1 - \bar{x}_2}{s_{\text{pooled}}}$$

Conventional benchmarks:

|d|	Classification
< 0.2	Negligible
0.2 – 0.5	Small
0.5 – 0.8	Medium
> 0.8	Large

- Winning ratio, thin vs. deep competition: d ≈ **0.96** (large — competition dominates)
- Unit price, winners vs. losers: d ≈ **–0.08** (negligible — and p = 0.087 agrees)

What Cohen's d Looks Like



The competition effect is the right-hand panel. Note how much the two curves **still overlap** at d = 0.96 — "large" does not mean separate, it means the centres have moved about one standard deviation apart. The winners-vs-losers price comparison, at d = –0.08, would be narrower than even the left-hand panel: two curves you could not tell apart.

Quick Check — Multiple t-Tests

60 seconds · pick A or B

You want to compare unit prices across the four line items in the extract. Instead of ANOVA, you run **6 pairwise t-tests** (all item pairs), each at $\alpha = 0.05$.

What is the actual probability of at least one false alarm across all 6 tests?

- **A)** 5% — same as each individual test
- **B)** Something higher — multiple tests inflate the error rate

✅ False Alarm Rate: ≈ 26%

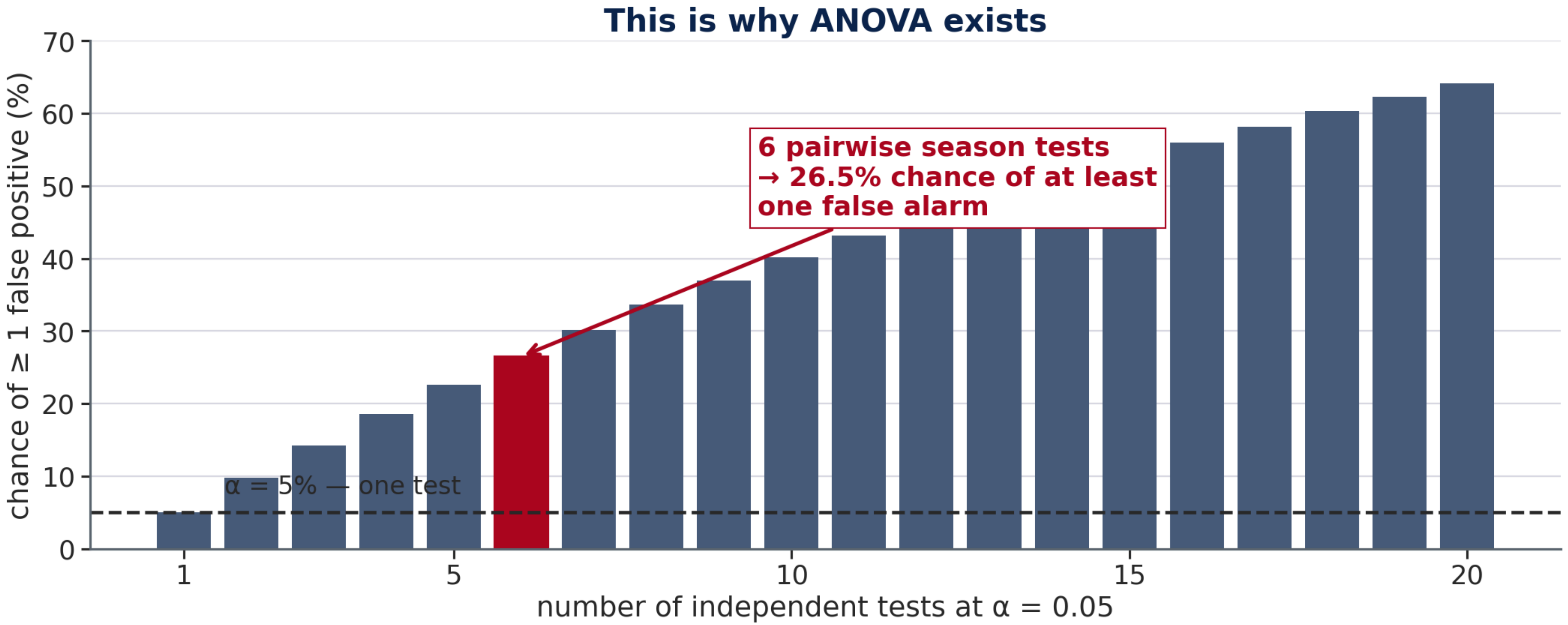
Running 6 t-tests at α = 0.05 is not α = 0.05

$$P(\text{at least one false alarm}) = 1 - (1-\alpha)^k = 1 - (0.95)^6 \approx \mathbf{26\%}$$

Tests run	Individual α	Family-wise false-alarm rate
1 test	5%	5%
3 tests	5%	14%
6 tests	**5%**	**26%**
10 tests	5%	40%

The fix: use **ANOVA** to first test "does any season differ?" (one test, one α = 0.05). If significant, use **Tukey HSD** for pairwise comparisons.

False Alarms Compound



Each test is honest on its own. Run enough of them and a false positive somewhere becomes the expected outcome rather than the exception.

One-Way ANOVA — Comparing Three or More Groups

The t-test compares **two** groups. When there are **k ≥ 3** groups (e.g., four seasons), use ANOVA.

H_0: All group means are equal (μ_Fall = μ_Spring = μ_Summer = μ_Winter)
H_1: At least one group mean differs

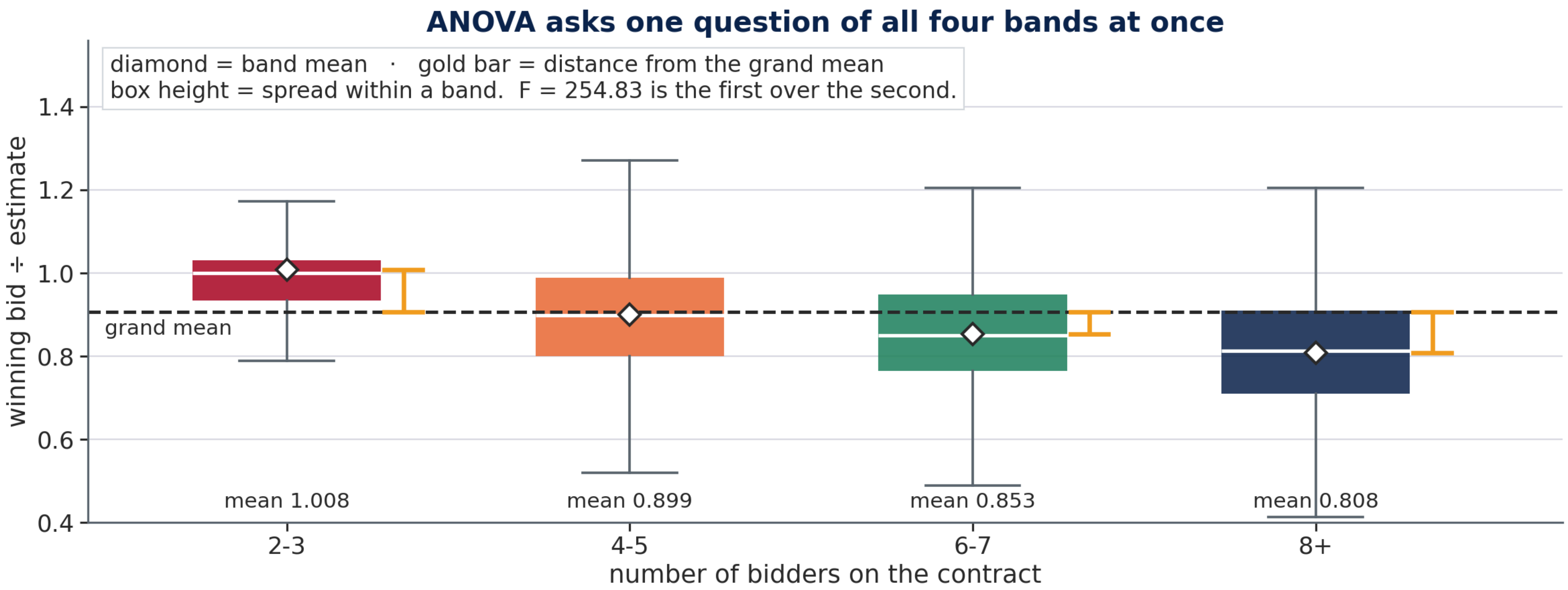
One-way ANOVA asks a single question of all four seasons at once: is *any* seasonal mean different from the others?

The **F-ratio** = between-group variance / within-group variance:

- Large F → the groups are far apart relative to the spread within each group → reject H_0
- $F \approx 1$ → the group means don't deviate more than within-group noise → do not reject H_0

Important limitation: ANOVA tells you *that* at least one group differs — not *which* groups. You need a **post-hoc test** to identify which specific pairs are different.

What ANOVA Is Comparing



F is a ratio: how far apart the group means are, divided by how spread out the data is inside each group. F = 254.83 says the competition signal dwarfs the contract-to-contract noise.

Post-Hoc Testing — Tukey HSD

After rejecting the ANOVA null hypothesis, **Tukey's Honestly Significant Difference (HSD)** test compares all pairs of groups while controlling the family-wise error rate.

Tukey's HSD follows the ANOVA and reports every pairwise comparison with its confidence interval, already corrected for the number of comparisons made.

The tutorial covers running it and reading the table.

The output table has one row per pair (e.g., Fall–Spring, Fall–Summer, …). Each row shows:

- `meandiff` — the estimated difference between group means
- `reject` — True if that pair is significantly different at $\alpha = 0.05$

Why not just run all six pairwise t-tests? Running multiple tests inflates the probability of a false alarm. With 6 tests at $\alpha = 0.05$, there is a 26% chance of at least one false positive. Tukey HSD adjusts for this.

Connection to Regression — t-Values in OLS Output

The `P>|t|` column in the `statsmodels` summary is doing a t-test on every coefficient:

H_0: that coefficient = 0 (the predictor has no linear effect on the outcome)
H_1: the coefficient ≠ 0

The regression output already carries a *t* statistic and *p*-value for **each coefficient** — a hypothesis test per predictor, computed when the model was fit.

Regressing $\log_{10}$ unit price on $\log_{10}$ quantity for excavation gives a slope of **–0.167** with **t ≈ –47.0**. Bigger jobs earn lower unit prices, and the effect is not close to zero.

On a log–log scale that slope is an elasticity: ten times the quantity buys about a **32% lower** unit price, since 10^(–0.167) ≈ 0.68.

When to Use Which Test

Situation	Test	Python function
Compare two group means	Two-sample t-test	`ttest_ind(g1, g2)`
Test a mean against a known value	One-sample t-test	`ttest_1samp(data, val)`
Compare three or more group means	One-way ANOVA	`f_oneway(g1, g2, g3, ...)`
Find which pairs differ after ANOVA	Tukey HSD	`pairwise_tukeyhsd(...)`
Measure practical size of an effect	Cohen's d	compute manually

Workflow: ANOVA first (does any group differ?) → if rejected, Tukey HSD (which pairs differ?) → report effect size (how large is the effect?).

Hypothesis Testing Across CE Practice

The test does not care what the numbers measure. Only the vocabulary changes.

Domain	Two groups being compared	The decision it informs
Construction mgmt	thin vs. deep competition	how to package and advertise work
Materials	cylinder breaks against a specified f′c	accept the pour, or core it
Transportation	crash rates before and after a retiming	keep the change, or revert it
Water resources	runoff on two catchment types	which design storm method to trust
Building construction	permit approval times by jurisdiction	how much float to carry

> 💡 The last row is this week's method for an architectural engineer — the same test, run on days instead of dollars.

Bridge to Machine Learning — Feature Selection

Hypothesis-testing concept	ML equivalent	Connection
Null hypothesis test	Feature importance test	Is this predictor's coefficient zero?
p-value threshold ($\alpha = 0.05$)	Feature selection filter	A significance cutoff becomes a keep/drop rule
Cohen's d	Normalized feature contribution	Effect size — a standardized difference
ANOVA F-test	Between-class variance ratio	Does this feature explain group variance?

In ML, **feature selection** often uses statistical tests under the hood — for example, selecting features by ANOVA F-score before feeding them to a model. The logic is identical to what you do this week: does this variable explain meaningful variance across groups?

Three Deliverables This Week

	What it covers	When
Weekly Quiz	15 questions on this lecture and the Tutorial	Closes 8:00 AM Tuesday, Nov 10
In-Class Exercise	The Colab notebook — all sections, plots, and written answers	Work Wed + Fri · due Wed 9:00 AM next week
Homework	A separate take-home problem set	On your own · due Wed 9:00 AM next week

The exercise spans two sessions. You start it Wednesday after the tutorial, and finish it Friday. **You are not expected to finish in class** — use the session time for the parts where having me in the room actually helps, and complete the rest on your own.

> 💡 Three submissions, three D2L Assignments folders. The quiz closes 8:00 AM Tuesday, Nov 10; the exercise and the homework are both due Wednesday 9:00 AM — no late work is accepted.

Looking Ahead — Week 12: Sensitivity Analysis

This week: *is an observed difference statistically real?*

Next week: *if the coefficients of a model are uncertain, how much uncertainty does that create in the answer?*

Sensitivity analysis identifies which input drives the most output variation. You will apply it to the bid-price model from this week's in-class exercise:

- If the quantity–price slope is off by 10%, how much does a cost forecast move?
- Does the district a job sits in matter more than the size of the job?
- Which of those deserves the most careful estimating effort before you sign?

This is a core engineering skill: knowing where to invest effort in reducing uncertainty.